BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

Section-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** VERSION: A

Note: There are three sections in this paper i.e. A, B & C. Instructions: Please attempt Section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, over writing, erasing, or multiple circles shading.

Q1. Choose the correct option i.e. A, B,C, or D.

Q1. 10^{-12} stands for .....

(A) Micro
(B) Pico
(C) Femto
(D) Nano

Q2. Which of the following is a vector quantity?

(A) Speed
(B) Distance
(C) Displacement
(D) Power

Q3. A train is moving at a speed of 36 kmh^{-1}. Its speed expressed in ms^{-1} is .....

(A) 10 ms^{-1}
(B) 20 ms^{-1}
(C) 25 ms^{-1}
(D) 30 ms^{-1}

Q4. The centripetal acceleration for an object of mass 1 kg moving with 6 ms^{-1} in a circle of radius 3 m is .....

(A) 18 ms^{-2}
(B) 12 ms^{-2}
(C) 10 ms^{-2}
(D) 2 ms^{-2}

Q5. Rolling friction is always ..... the sliding friction.

(A) Greater than
(B) Equal to
(C) Less than
(D) Both A and C

Q6. Conventionally clock wise torque is taken as .....

(A) Negative
(B) Positive
(C) Parallel
(D) Zero

Q7. The angle at which x and y components of force are equal is .....

(A) 0°
(B) 30°
(C) 45°
(D) 60°

Q8. At Earth's surface, Earth pulls down on a 1 kg mass with a force of magnitude .....

(A) 9.8 N
(B) 98 N
(C) 0.9 N
(D) 0.09 N

Q9. One horse power (hp) = ...

(A) 74.6 Watt
(B) 7.46 Watt
(C) 0.746 Watt
(D) 746 Watt

Q10. The unit of Young's Modulus is same as that of .....

(A) Strain
(B) Power
(C) Stress
(D) Force

Q11. The amount of heat required to raise the temperature of 1 kg of water by 1 K is .....

(A) 1 J
(B) 400 J
(C) 310 J
(D) 4190 J

Q12. The mode of transfer of heat which does not require material medium is called .....

(A) Convection
(B) Conduction
(C) Radiation
(D) None

Section-B

Time Allowed: 2:40 Hours | **Maximum Marks:** 32

Q2. Write short answers to any eight (8) of the following parts. All parts carry equal marks.

(i) What are the significant figures in the following measurement: (a) 1009 m (b) 0.00450 kg (c) 1.66×10^{-27} kg (d) 2001 sec

(ii) Is it possible that displacement is zero but not the distance, under what condition displacement will be equal to distance?

(iii) An iron shot of mass 6 g is fired with an airgun. If the velocity of the shot is 62 ms^{-1}. What is the magnitude of momentum?

(iv) How does friction help you walk? Is it kinetic friction or static friction?

(v) State first and second conditions of equilibrium.

(vi) Write a note on Law of Universal Gravitation.

(vii) What would happen to your weight on earth if the mass of the earth doubled, but its radius stayed the same?

(viii) A 60 g bullet is fired from a gun with 3150 J of kinetic energy. Find its velocity.

(ix) Why a small needle sinks in water and huge ships travel easily in water without sinking?

(x) Explain Anomalous expansion of water.

(xi) Explain why burns caused by steam at 100 °C on the skin are often more severe than burns caused by water at 100 °C.

Section-C

Maximum Marks: 21

> Note: Answer any three (3) questions. All questions carry equal marks.

Q3.

(a) [4] What is isolated system? State law of conservation of momentum. Explain law of conservation of momentum in firing of rifle.

(b) [3] Divia is pulling a box on the floor with a force of 20 N making an angle of 60° with the horizontal. Find the horizontal and vertical components of this force.

Q4.

(a) [4] Show that the orbital speed depends upon the mass of earth and the distance from the center of mass of the satellite.

(b) [3] What power is required for a ski-hill chair lift that transports 500 people (average mass 65 kg) per hour to an increased elevation of 1200 m?

Q5.

(a) [4] Explain Pascal's Principle.

(b) [3] A rectangular glass block of dimensions 30 cm by 5 cm by 10 cm weighs 37.5 N. Calculate the least and the greatest pressure it can exert when resting on a horizontal table?

Q6.

(a) [4] Explain volume thermal expansion of solids. Also define co-efficient of volume thermal expansion and what is its unit?

(b) [3] What is the specific heat of a metal substance if 135 kJ of heat is needed to raise 4.1 kg of the metal from 18.0 °C to 37.2 °C?